

A3-PS019-TC057

Personal-Data Discovery and Privacy Evidence Inventory

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Problem Description

Personal data piles up in unindexed tables and spreadsheets, detached from any declared purpose or retention schedule.

- PII spreads across CSV exports and SQL tables without documentation
- Purpose and retention metadata is missing or not linked to the fields
- Reviewers have no technical evidence inventory to work from

Challenge: a scanner for synthetic CSV files and one SQL database that finds candidate personal-data fields in five data classes, links them to purpose and retention metadata, and flags gaps. It produces evidence for human review and makes no legal compliance decisions.

Background

Data protection frameworks expect organisations to know what personal data they hold, why, and for how long.

- India's DPDP Act, 2023 and the MeitY framework tie processing to a stated purpose and limited retention [1, 2]
- Records of processing (purpose, retention, data categories) are the usual way to show this [3, 4]
- Detection is probabilistic: a scanner finds *candidate* fields, a human confirms them
- Technical evidence (what was found, where) is kept separate from legal interpretation

Literature & RFC Review

Only the most relevant laws, standards and tools are reviewed.

- **DPDP Act 2023, MeitY framework** [1, 2]: purpose-bound processing and storage limitation
- **GDPR Art. 30, ISO/IEC 27701** [3, 4]: model for the inventory fields (purpose, retention, categories)
- **Microsoft Presidio** [5]: open-source PII detection with configurable recognizers, which informs our rule design
- **Verhoeff checksum** [6]: validates Aadhaar-style numbers and cuts false positives

Proposed Solution

A configurable scanner that turns CSV and SQL schemas into a privacy evidence inventory.

1. **Connectors** read CSV files and one SQL database (schema plus sampled values)
2. **Ruleengine**(YAML):column-namepatterns,valueregexesandchecksumvalidators

   produce candidate fields with evidence and a confidence score
3. **Metadata mapper** links fields to a purpose and retention register
4. **Gapreport**flagsfieldswithmissingpurposeorretention
5. **Evaluator**reportsper-classprecisionandrecallonamanuallylabelledtestset

Five classes (configurable): name, email, phone, government ID (Aadhaar/PAN), date of birth.
Out of scope: lawful-basis determination and universal data-flow inference.

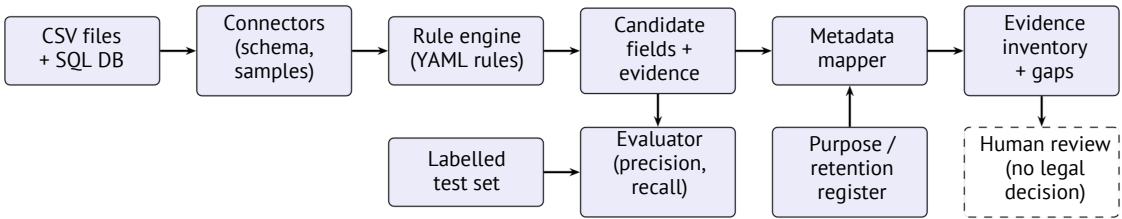
Optimization Proposed

Planned improvements over a plain column-name search.

Aspect	Before	Proposed
Detection	Column names only	Names plus sampled values and validators
Rules	Hard-coded in script	Editable YAML, no code change
False positives	Any 12-digit column flagged	Checksum validation before flagging
Metadata	Manual spreadsheet lookup	Automatic mapping and gap flags
Output	Evidence mixed with opinion	Evidence only, reviewer notes kept apart

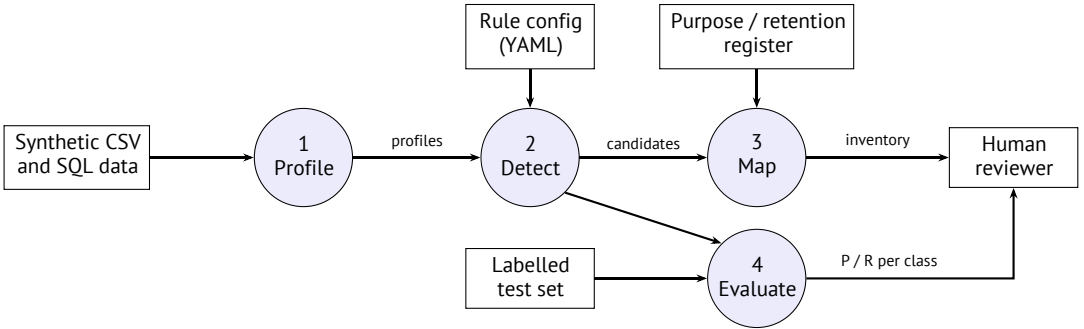
Solution Architecture & Design

Visual: Flowchart showing the scanning and evidence process.



Data Flow Diagram

Overview of the data flows between the scanner components.



Deployment Timeline

Iterative and evaluation-driven: each phase ends with a demonstrable result.

Phase 1	Private repo, synthetic CSV/SQL data, labelled test set
Phase 2	Connectors and configurable rule engine for five classes
Phase 3	Purpose/retention mapping and missing-metadata gap repor t
Phase 4	Per-class precision/recall, rule tuning
Phase 5	Final report, PDF in repository, demo with pseudocode

Bibliography I

[1] GovernmentofIndia.*TheDigitalPersonalDataProtectionAct,2023*(No.22of2023).

[2] MeitY.*DataProtectionFramework*.meity.gov.in/data-protection-framework.

[3] EuropeanUnion.*Regulation(EU)2016/679(GDPR)*,Art.30.

[4] ISO/IEC27701:2019.*PrivacyinformationmanagementextensiontoISO/IEC27001and27002*.

[5] Microsoft. *Presidio: Data Protection and De-identification SDK*. github.com/microsoft/presidio.

[6] J.Verhoeff.*ErrorDetectingDecimalCodes*.MathematicalCentreTract29,Amsterdam,1969.

Thank You for Your Attention!